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Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

Authors:

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\\huge Fluid compressibility, c_f

1. Introduction

The fluids contained within the porous media of petroleum reservoirs consist primarily of oil, water, and natural gas. Their distribution within the pore space depends on the thermodynamic conditions of the system, the petrophysical properties of the reservoir rock, and the physicochemical properties of the fluids. Fluid distribution may also change throughout the life of a reservoir as a result of natural depletion mechanisms or engineered operations, such as water injection, gas injection, or other enhanced recovery processes.

Understanding reservoir-fluid properties is essential for developing appropriate fluid-flow models. These models describe the physical response of reservoir fluids to changes in pressure and temperature. Consequently, the mathematical complexity and solution approach depend on the characteristics of the reservoir system, the number and type of fluid phases present, their interactions, and the flow regime being considered.

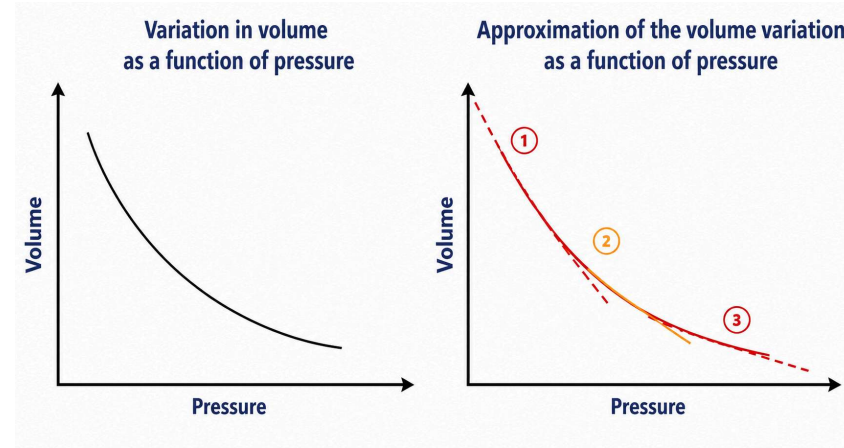
In dynamic reservoir characterization, fluid behavior plays a fundamental role because the relative volumes and distribution of the different phases directly influence fluid mobility, reservoir performance, recoverable volumes, and production potential. A proper description of these properties is therefore necessary to evaluate reservoir-drive mechanisms and to design appropriate production and recovery strategies.

Among the properties used to characterize reservoir fluids, volume change is particularly important. As reservoir conditions change, fluids may expand or contract, affecting their density, mobility, and the amount of fluid occupying the reservoir pore space. The magnitude of this volumetric response to pressure is quantified through fluid compressibility, which is one of the fundamental properties considered in reservoir engineering.

For the analysis presented in this notebook, the reservoir process is assumed to be isothermal. Therefore, temperature remains constant and changes in fluid volume are considered to result exclusively from variations in reservoir pressure. The fluid compressibility, c_f , is mathematically defined as follows:

$$c_f = -\frac{1}{V_{fluid}} \left(\frac{\partial V_{fluid}}{\partial p} \right)_T$$

The following figure shows the typical variation of the fluid compressibility as a function of the reservoir pressure.



- **Incompressible fluid.** An incompressible fluid is a fluid that experiences negligible or no change in volume when subjected to variations in pressure. Therefore, the derivative of fluid volume with respect to pressure is approximately zero:

$$\frac{\partial V_{fluid}}{\partial p} \approx 0$$

and consequently, the fluid compressibility, c , is also approximately zero:

$$c_f = -\frac{1}{V_{fluid}} \left(\frac{\partial V_{fluid}}{\partial p} \right)_T \approx 0$$

The incompressible-fluid assumption is commonly applied to the water phase in reservoir engineering because formation water typically exhibits relatively small volumetric changes over moderate pressure ranges. As a result, many analytical models and numerical reservoir simulations treat water as incompressible without introducing significant errors for numerous practical applications. This assumption can considerably simplify the mathematical formulation of fluid-flow problems, since changes in fluid density and volume with pressure can be neglected.

However, formation water is not truly incompressible. It has a small but finite compressibility, and its volume and density change slightly with pressure. Therefore, the validity of the incompressible-fluid assumption depends on the specific reservoir problem, pressure range, required level of accuracy, and quality of the available PVT data. For problems involving large pressure variations, highly accurate material-balance calculations, pressure-transient analysis, or detailed reservoir simulation, water compressibility may need to be explicitly included rather than neglected.

- **Slightly compressible fluid.** In the development of many theoretical models involving liquid flow through porous media, it is commonly assumed that the fluid compressibility remains constant within a specified pressure range. This assumption is particularly useful for liquids because their compressibility is generally small and varies only moderately over limited pressure intervals.

Under this assumption:

$$c_f = -\frac{1}{V_{fluid}} \left(\frac{\partial V_{fluid}}{\partial p} \right)_T = \text{constant}$$

A fluid that exhibits a small but nonzero compressibility that can reasonably be considered constant is commonly referred to as a slightly compressible fluid.

Assuming constant compressibility reduces the number of pressure-dependent variables involved in the model and considerably simplifies its mathematical formulation and numerical solution. This approximation is widely used in reservoir engineering and fluid-flow models, particularly for liquids over pressure ranges where changes in fluid properties remain relatively small. However, the validity of this assumption depends on the fluid properties, pressure range, and required accuracy of the model. Over large pressure intervals or when fluid properties vary significantly, compressibility should be treated as a function of pressure rather than as a constant.

When fluid compressibility is assumed to be constant, the variation of fluid volume with pressure can be represented using a relatively simple relationship that adequately approximates fluid behavior over the pressure interval of interest. This assumption is widely used in fluid-flow calculations and reservoir simulation, particularly for the oil phase when it behaves as a slightly compressible fluid.

For relatively small pressure changes and small values of compressibility, the fluid compressibility can be approximated by a linear expression:

$$c_f = V_{ref} [1 + c_f(p_{ref} - p)]$$

This linear approximation is particularly convenient in reservoir-engineering calculations because it provides a simple representation of the expansion or contraction of a slightly compressible fluid as reservoir pressure changes.

- **Compressible fluid.** Fluids in this classification exhibit a strong dependence of volume on pressure. Consequently, their compressibility cannot generally be assumed to remain constant over the pressure range of interest and must instead be treated as a pressure-dependent property. Reservoir gas is the most representative example of a compressible fluid. Unlike liquids, gases can undergo substantial changes in volume and density in response to relatively moderate changes in pressure. As reservoir pressure decreases, gas expands significantly; conversely, increasing pressure causes considerable gas compression.

At constant temperature and for a fixed amount of gas, the variation of gas volume with pressure therefore depends not only on p, but also on the variation of Z with pressure. As a result, the isothermal gas compressibility, c_g , can be expressed as:

$$c_g = \frac{1}{p} - \frac{1}{z} \left(\frac{\partial z}{\partial p} \right)_T$$

2. Numerical solution

Incompressible fluid

$$c_f = constant$$

Slightly compressible fluid

$$V = V_{ref} [1 + c_f(p_{ref} - p)]$$

Compressible fluid

$$c_g = \frac{1}{p} - \frac{1}{z} \left(\frac{\partial z}{\partial p} \right)_T$$

2.1 Import libraries

```
In [2]: import numpy as np
import matplotlib.pyplot as plt
```

2.2 Inputs

```
In [43]: cfref=5*10**-6

Vref=200
pref=100

p=np.linspace(100,20000,100)
```

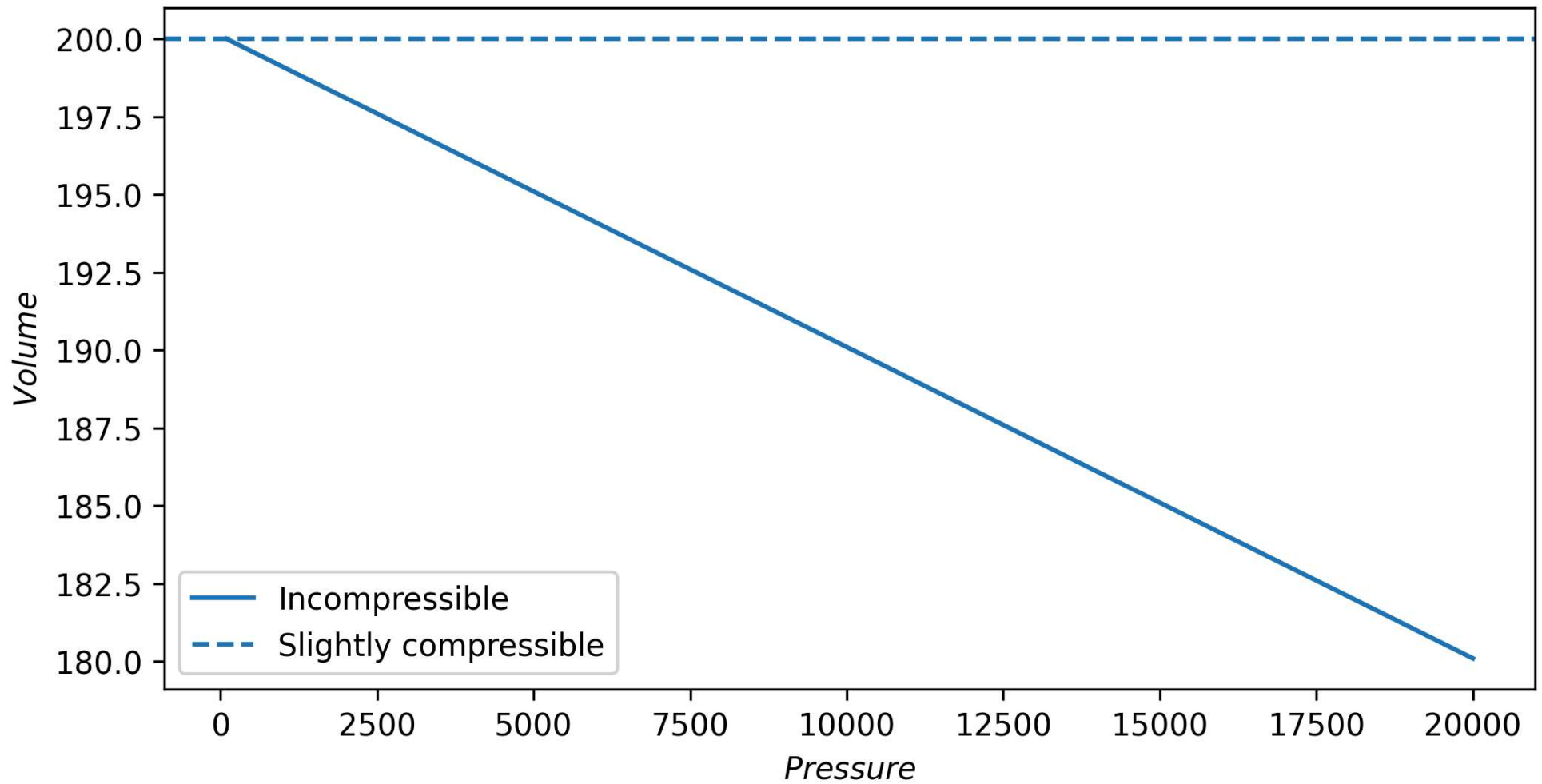
2.3 Calculation and plotting

```
In [44]: plt.figure(dpi=300, figsize=(8,4))

V=Vref*(1+cfref*(pref-p))

plt.plot(p,V, label='Incompressible')
plt.axhline(y=Vref, linestyle='--', label='Slightly compressible')
plt.xlabel('$Pressure$')
plt.ylabel('$Volume$')
```

```
# plt.grid(True)
plt.legend()
plt.show()
```



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

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In []: